

Questions with Answer Keys

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Q1 - Shifting of origin and Rotation of axis

If the axes are rotated through an angle of 30° in the anti-clockwise direction, then coordinates of point

$(4, -2\sqrt{3})$ with respect to new axes are

(1) $(2\sqrt{3})$

(2) $(\sqrt{3}, -5)$

(3) $(2, 3)$

(4) $(\sqrt{3}, 2)$

Q2 - Shifting of origin and Rotation of axis

Reflecting the point $(2, -1)$ about Y -axis, coordinate axes are rotated at 45° angle in negative direction without shifting the origin. The new coordinates of the points are

(1) $\left(-\frac{1}{\sqrt{2}}, -\frac{3}{\sqrt{2}}\right)$

(2) $\left(\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$

(3) $\left(-\frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

(4) None of these

Q3 - Shifting of origin and Rotation of axis

Without changing the direction of coordinate axes, to which point should origin be transferred, so that the

equation $x^2 + y^2 - 4x + 6y - 7 = 0$ is changed to an equation which contains no term of first degree?

(1) $(3, 2)$

(2) $(2, -3)$

(3) $(-2, 3)$

(4) None of these

Q4 - Angle between Lines, Image and Foot of Perpendicular

If $x - 3y = p$, $ax + 2y = q$ and $ax + y = r$ form a right angled triangle, then

Questions with Answer Keys

#MathBoleTohMathonGo

(1) $a^2 - 6a - 12 = 0$

(2) $a^2 - 9a + 12 = 0$

(3) $a^2 - 9a + 18 = 0$

(4) $a^2 - 6a - 18 = 0$

Q5 - Angle between Lines, Image and Foot of Perpendicular

The coordinates of foot of a perpendicular drawn on line $3x - 4y - 5 = 0$ to the point $(0, 5)$, is

(1) $(1, 3)$

(2) $(2, 3)$

(3) $(3, 2)$

(4) $(3, 1)$

Q6 - Angle between Lines, Image and Foot of Perpendicular

The ratio in which the line $3x + 4y + 2 = 0$ divides the distance between $3x + 4y + 5 = 0$ and

$3x + 4y - 5 = 0$, is

(1) $7 : 3$

(2) $3 : 7$

(3) $2 : 3$

(4) None of these

Q7 - Angle between Lines, Image and Foot of Perpendicular

Let line L be $2x + y = 2$. If the coordinate axes be rotated at 45° angle, then the lengths of the intercepts of L on new axes will be

(1) $\sqrt{2}, 1$

(2) $1, \sqrt{2}$

(3) $2\sqrt{2}, 2\sqrt{\frac{2}{3}}$

(4) $\frac{2\sqrt{2}}{3}, 2\sqrt{2}$

Questions with Answer Keys

#MathBoleTohMathonGo

Q8 - Angle between Lines, Image and Foot of Perpendicular

The reflection of the point (3,8) with respect to the line $x + 3y = 7$, is

- (1) (5,-6)
- (2) (-1,-4)
- (3) (0,-1)
- (4) (-9,-4)

Q9 - Straight Line Making Given Angle with Line

A vertex of an equilateral triangle is (2,3) and the equation of the opposite side is $x + y = 2$, then the equation of other two sides of the triangle, are

- (1) $(y - 3) = (2 \pm \sqrt{3})(x - 2)$
- (2) $(y - 3) = (4 \pm \sqrt{3})(x + 2)$
- (3) $(y - 3) = (2 \pm \sqrt{3})(x + 4)$
- (4) $(y - 3) = (6 \pm \sqrt{3})(x - 2)$

Q10 - Straight Line Making Given Angle with Line

(1,2) is vertex of a square whose one diagonal is along the X -axis. The equations of sides passing through the given vertex are

- (1) $2x - y = 0, x + 2y + 5 = 0$
- (2) $x - 2y + 3 = 0, 2x + y - 4 = 0$
- (3) $x - y + 1 = 0, x + y - 3 = 0$
- (4) None of these

Q11 - Straight Line Making Given Angle with Line

The diagonal of the parallelogram whose sides are $lx + my + n = 0$

$lx + my + n' = 0, mx + ly + n = 0, mx + ly + n' = 0$ include an angle

- (1) $\tan^{-1} \left(\frac{2lm}{l^2 + m^2} \right)$

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Questions with Answer Keys

#MathBoleTohMathonGo

$$(2) \tan^{-1} \left(\frac{l^2 - m^2}{l^2 + m^2} \right)$$

$$(3) \frac{\pi}{2}$$

$$(4) \frac{\pi}{3}$$

Q12 - Angle Bisector and Family of lines

The equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$, is

$$(1) 11x - 3y - 9 = 0$$

$$(2) 11x - 3y + 9 = 0$$

$$(3) 21x + 77y - 101 = 0$$

(4) None of these

Q13 - Angle Bisector and Family of lines

If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}, x > 0$, and $y = 3x, x > 0$ then a belongs to

$$(1) (3, \infty)$$

$$(2) \left(\frac{1}{2}, 3 \right)$$

$$(3) \left(-3, -\frac{1}{2} \right)$$

$$(4) \left(0, \frac{1}{2} \right)$$

Q14 - Angle Bisector and Family of lines

The straight line $ax + by + c = 0$, where $3a + 2b + 4c = 0$ are concurrent at the point

$$(1) \left(\frac{1}{2}, \frac{3}{4} \right)$$

$$(2) \left(\frac{3}{4}, \frac{1}{2} \right)$$

$$(3) \left(-\frac{3}{4}, -\frac{1}{2} \right)$$

$$(4) \left(-\frac{3}{4}, \frac{1}{2} \right)$$

Q15 - Angle Bisector and Family of lines

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Questions with Answer Keys

#MathBoleTohMathonGo

For the straight lines given by the equation $(2 + k)x + (1 + k)y = 5 + 7k$, for different values of k . Which of the following statements is true?

- (1) Lines pass through the point $(2, -9)$
- (2) Lines pass through the point $(-2, 9)$
- (3) Lines are parallel
- (4) None of the above

Q16 - Angle Bisector and Family of lines

A is a point on either of two lines $y + \sqrt{3}|x| = 2$ at a distance of $\frac{4}{\sqrt{3}}$ units from their point of intersection. The coordinates of the foot of perpendicular from A on the bisector of the angle between them are

- (1) $\left(\frac{-2}{\sqrt{3}}, 2\right)$
- (2) $(0, 0)$
- (3) $\left(\frac{2}{\sqrt{3}}, 2\right)$
- (4) $(0, 4)$

Q17 - Angle Bisector and Family of lines

In a $\triangle ABC$, if $A(2, -1)$ and $7x - 10y + 1 = 0$ and $3x - 2y + 5 = 0$ are equations of an altitude and an angle bisector respectively drawn from B , then equation of BC is

- (1) $x + y + 1 = 0$
- (2) $5x + y + 17 = 0$
- (3) $4x + 9y + 30 = 0$
- (4) $x - 5y - 7 = 0$

Q18 - Locus

For variable lines $\frac{x}{a} + \frac{y}{b} = 1$, $a + b = 10$, the locus of mid-point of the intercept of this line between coordinate axes is

- (1) $10x + 5y = 1$

Questions with Answer Keys

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(2) $x + y = 10$

(3) $x + y = 5$

(4) $5x + 10y = 1$

Q19 - Locus

If $A(2, -3)$ and $B(-2, 1)$ are vertices of a $\triangle ABC$ and centroid of the triangle lies on the line $2x + 3y = 1$, then locus of the vertex C , is

(1) $3x + 2y = 5$

(2) $2x - 3y = 7$

(3) $2x + 3y = 9$

(4) $3x - 2y = 3$

Q20 - Locus

Vertices of a variable triangle are $A(3, 4)$, $B(5 \cos \theta, 5 \sin \theta)$ and $C(5 \sin \theta, -5 \cos \theta)$. Then, locus of its orthocentre is

(1) $(x + y - 1)^2 + (x - y - 7)^2 = 100$

(2) $(x + y - 7)^2 + (x - y - 1)^2 = 100$

(3) $(x + y - 7)^2 + (x + y + 1)^2 = 100$

(4) $(x + y - 7)^2 + (x - y + 1)^2 = 100$

Q21 - Locus

If the distance of any point (x, y) from the origin is defined as $d(x, y) = \max\{|x|, |y|\}$, then the locus of the point (x, y) , where $d(x, y) = 1$, is

(1) a square of area 1 sq unit

(2) a square of area 4 sq units

(3) a triangle

(4) None of these

Questions with Answer Keys

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Q22 - Pair of straight lines

If $3x + 4y = 0$ is one of the lines represented by $6x^2 - xy + 4cy^2 = 0$, then c is equal to

- (1) 3
- (2) -1
- (3) 1
- (4) -3

Q23 - Pair of straight lines

If slope of lines represented by $ax^2 + 2hxy + by^2 = 0$ are in the ratio 1 : 3, then $h^2 : ab$ equals

- (1) $\frac{3}{4}$
- (2) $\frac{1}{4}$
- (3) $\frac{4}{3}$
- (4) None of these

Q24 - Pair of straight lines

If one of the pair of lines $ax^2 + 2hxy + by^2 = 0$ bisects the angle between coordinates axes in positive quadrant, then

- (1) $a - b = 2|h|$
- (2) $a + b = -2h$
- (3) $a + b = 2|h|$
- (4) $(a - b)^2 = 4h^2$

Q25 - Pair of straight lines

The nature of straight lines represented by equation $4x^2 + 12xy + 9y^2 = 0$, is

- (1) real and coincident
- (2) real and different
- (3) imaginary and different

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Questions with Answer Keys

#MathBoleTohMathonGo

(4) None of the above

Q26 - Pair of straight lines

The lines $ax^2 + bcy^2 = a(b + c)xy$ will be coincident, if(1) $a = 0$ or $b = c$ (2) $a = b$ or $a = c$ (3) $c = 0$ or $a = b$ (4) $a = b + c$

Q27 - Pair of straight lines

If the lines represented by the equation $ax^2 - bxy - y^2 = 0$ make angles α and β with the X -axis, then $\tan(\alpha + \beta)$ is equal to(1) $\frac{b}{(1+a)}$ (2) $\frac{-b}{(1+a)}$ (3) $\frac{a}{(1+b)}$

(4) None of these

Q28 - Pair of straight lines

If the lines $(p - q)x^2 + 2(p + q)xy + (q - p)y^2 = 0$ are mutually perpendicular, then(1) $p = q$ (2) $p = 0$ (3) $q = 0$ (4) p and q may have any value

Q29 - Pair of straight lines

If the bisectors of angles represented by $ax^2 + 2hxy + by^2 = 0$ and $a'x^2 + 2h'xy + b'y^2 = 0$ are same, then(1) $(a - b)h' = (a' - b')h$

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Questions with Answer Keys

#MathBoleTohMathonGo

$$(2) (a - b)h = (a' - b')h'$$

$$(3) (a + b)h' = (a' - b')h$$

$$(4) (a - b)h' = (a' + b')h$$

Q30 - Homogenization of curves

The equation of pair of straight lines joining the point of intersection of the curve $x^2 + y^2 = 4$ and $y - x = 2$ to the origin, is

$$(1) x^2 + y^2 = (y - x)^2$$

$$(2) x^2 + y^2 + (y - x)^2 = 0$$

$$(3) x^2 + y^2 = 4(y - x)^2$$

$$(4) x^2 + y^2 + 4(y - x)^2 = 0$$

Q31 - Homogenization of curves

The lines joining the points of intersection of the curve $(x - k)^2 + (y - h)^2 = c^2$ and the line $hx + ky = 2hk$ to the origin are perpendicular, then

$$(1) c = h \pm k$$

$$(2) c^2 = h^2 + k^2$$

$$(3) c^2 = (h + k)^2$$

$$(4) 4c^2 = h^2 + k^2$$

Q32 - Homogenization of curves

The lines joining the origin to the points of intersection of the line $y = mx + c$ and the circle $x^2 + y^2 = a^2$ will be mutually perpendicular, if

$$(1) a^2(m^2 + 1) = c^2$$

$$(2) a^2(m^2 - 1) = c^2$$

$$(3) a^2(m^2 + 1) = 2c^2$$

$$(4) a^2(m^2 - 1) = 2c^2$$

Questions with Answer Keys

#MathBoleTohMathonGo

Answer Key

Q1 (2)	Q2 (1)	Q3 (2)	Q4 (3)
Q5 (4)	Q6 (2)	Q7 (4)	Q8 (2)
Q9 (1)	Q10 (3)	Q11 (3)	Q12 (2)
Q13 (2)	Q14 (2)	Q15 (2)	Q16 (2)
Q17 (2)	Q18 (3)	Q19 (3)	Q20 (4)
Q21 (2)	Q22 (4)	Q23 (3)	Q24 (2)
Q25 (1)	Q26 (1)	Q27 (2)	Q28 (4)
Q29 (1)	Q30 (1)	Q31 (2)	Q32 (3)